Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Recovering axial structure from fixed-focus fluorescence images requires measurements that distinguish different depth arrangements. We learn depth-encoding illumination from the optical model alone, without specimen examples or a reconstruction decoder. An angle-scanned plane wave passes through a fixed, defocused phase plate, generating depth-dependent excitation while the specimen and detection focus remain stationary. The plate is optimized by reducing weighted squared correlations between detector-integrated measurement signatures at different depths and the same lateral position. We evaluate the resulting optics using total variation, two plug-and-play methods and a feed-forward decoder. On simulated mouse pyramidal neuron volumes at eightfold measurement compression and matched nominal excitation dose, the same plate reduces axial-profile Wasserstein-1 (W1) error by 27–50 % relative to its untrained initialization. Under the three independent reconstruction methods, a jointly trained plate's mean W1 differs from ours by 1.9–3.2 %. With its own decoder, the end-to-end system achieves 12 % lower W1. Our criterion also improves upon the tested Fisher-information and worst-case-coherence designs. Measurement-level tests show stronger discrimination of prescribed axial alternatives, and low-dose crossings reveal a structural benefit over the tested joint design. The optical model alone can thus guide illumination design for axial recovery across the tested reconstruction methods.

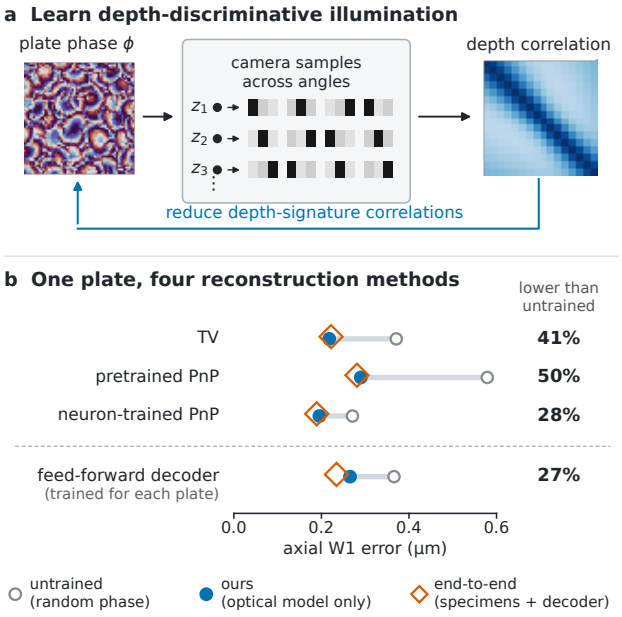


Figure 1. **Decoder-free optical learning improves axial recovery across reconstruction methods.** (a) Training decorrelates detector-integrated depth signatures; strips are schematic. (b) Mean W1 on simulated measurements at $C = 8$ and matched nominal dose. Percentages compare ours with the shared untrained initialization. Legend inputs refer to optical training. Feed-forward networks are fine-tuned for each plate, including the end-to-end plate. Paired intervals are in the supplement.

1. Introduction

Fluorescent branches that overlap in a camera image may lie at different depths. Recovering their axial arrangement requires measurements that distinguish those depths, particularly when detection has a broad depth of field. If different arrangements produce similar observations, reconstruction relies more heavily on prior assumptions [2, 7]. An accurate mean depth can still conceal two structures reconstructed as one. We therefore consider the full axial intensity profile at each lateral position and ask: can learning the illumination improve its recovery without choosing a reconstruction method during optical training?

Our system scans the incident angle of a plane wave through a fixed, defocused phase plate (Fig. 2). Propagation produces structured excitation whose lateral shift under the scan varies with depth. Structures at different axial positions therefore experience different sequences of excitation intensities. A fixed-focus camera records one fluorescence image per angle while the plate, specimen and detection focus remain stationary. The plate is the only learned optical element; after training, the same phase map is evaluated with each reconstruction method.

The design question is which measurement responses the plate should produce. End-to-end optical learning optimizes

an optical–decoder combination using specimen examples [4, 15]. A fixed optical element, however, can be reused as reconstruction methods change. Fisher-information and coherence-based designs already optimize optical responses independently of a decoder [21, 25]. We follow this principle for angle-scanned excitation. At each lateral position, a voxel's signature collects its camera response across illumination angles, including excitation, emission blur and pixel integration. We minimize weighted squared correlations between these signatures at different depths, using the optical model alone (Fig. 1a).

We test the design at both measurement and reconstruction stages. Simulated mouse pyramidal neuron volumes are compared with prescribed axial alternatives before reconstruction, then recovered using methods with different prior assumptions. Untrained encoders, alternative optical criteria and a jointly trained plate distinguish the effects of optical design from those of reconstruction (Fig. 1b).

Our contributions are:

1. **Depth encoded in excitation, learned without a decoder.** We develop angle-scanned excitation through a fixed, defocused phase plate and learn its phase by decorrelating detector-integrated depth signatures at each lateral position. Optical training requires neither specimen examples nor a reconstruction solver.
2. **Recovery gains across reconstruction methods.** At eightfold measurement compression, the learned plate lowers axial-profile error by 27–50 % relative to its untrained initialization under four reconstruction methods. It also gives lower error than the tested Fisher-information and worst-case-coherence designs under each method.
3. **Comparison with joint optical–decoder training.** Under the three independent reconstruction methods, mean W1 differs by 1.9–3.2 % between the two learned plates, favoring ours under TV and end-to-end under both PnP methods. With a separately fine-tuned feed-forward decoder for each plate, the end-to-end system gives 12 % lower error.

2. Related work

Depth encoding in fluorescence microscopy. Computational fluorescence microscopes encode depth through detection or excitation. Light-field microscopy, DiffuserCam and Miniscope3D recover volumes from depth-dependent detection responses [1, 3, 25]. Excitation-side methods use illumination that varies with depth. Yeh *et al*. [26] combine translated speckles with axial scanning, while Pascucci *et al*. [16] exploit axial speckle decorrelation for volumetric recovery from lateral scans, with saturation improving resolution. DEEP and DEEP^2 combine excitation patterning with computational fluorescence recovery through scattering [23, 29]. Angular scanning through a scattering layer enables two-dimensional fluorescence recovery [27], and tilted pattern projection produces depth-dependent lateral shifts [9]. Our configuration learns a defocused phase plate for angular excitation scanning, with the specimen and detection focus stationary.

Learning optical encoders. End-to-end methods optimize optics together with reconstruction networks for depth estimation and fluorescence imaging [4, 8, 15, 24]. Kellman *et al*. [11] optimize illumination through a specified reconstruction algorithm. Decoder-independent design instead targets properties of the optical responses. Shechtman *et al*. [21] optimize Fisher information for emitter localization; Miniscope3D [25] penalizes coherence between depth-dependent detection PSFs. Gram-matrix objectives also appear in sensing-matrix design [5, 6]. IDEAL optimizes estimated mutual information using measurement-distribution models [18]; IDEAL-IO alternates density fitting and optical optimization, including for snapshot 3D fluorescence microscopy [14]. We specialize coherence-based design to detector-integrated depth signatures across excitation angles, without specimen examples or a fitted specimen distribution. Our comparison tests both alternative model-only criteria and an end-to-end counterpart using the same optical model.

Volumetric reconstruction and priors. TV promotes piecewise-smooth structure [20], whereas PnP methods use denoisers as implicit priors [10, 22]. Multi-slice fusion combines two-dimensional denoisers along complementary orientations of a shared volume [12]. We adapt this framework to fluorescence reconstruction and evaluate the optics with TV, pretrained and neuron-trained PnP, and a feed-forward learned decoder. This separates optical training from the prior choices used to assess recovery.

3. Method

Depth from the scan angle. A tilted plane wave illuminates a fixed phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$. Propagation converts its phase structure into spatially varying excitation. Changing the incident angle shifts this pattern differently at different depths. In the ideal paraxial thin-screen model, a transverse direction-cosine change $\Delta\mathbf{s}$ produces displacement $\Delta\mathbf{r} \approx d\,\Delta\mathbf{s}$ after distance d. Each depth therefore experiences a different excitation sequence during the scan. Plate defocus places the shear pivot outside the specimen. The high-NA vectorial relay and finite pupil determine the actual patterns, including departures from pure translation. Acquisition is sequential: one camera image is recorded per angle at a fixed detection focus (Fig. 2a).

Measurement model and signatures. Let $X_0(\mathbf{r}, z)$ denote the specimen's fluorescence response per unit excitation. Under linear, unsaturated fluorescence, illumination I_t

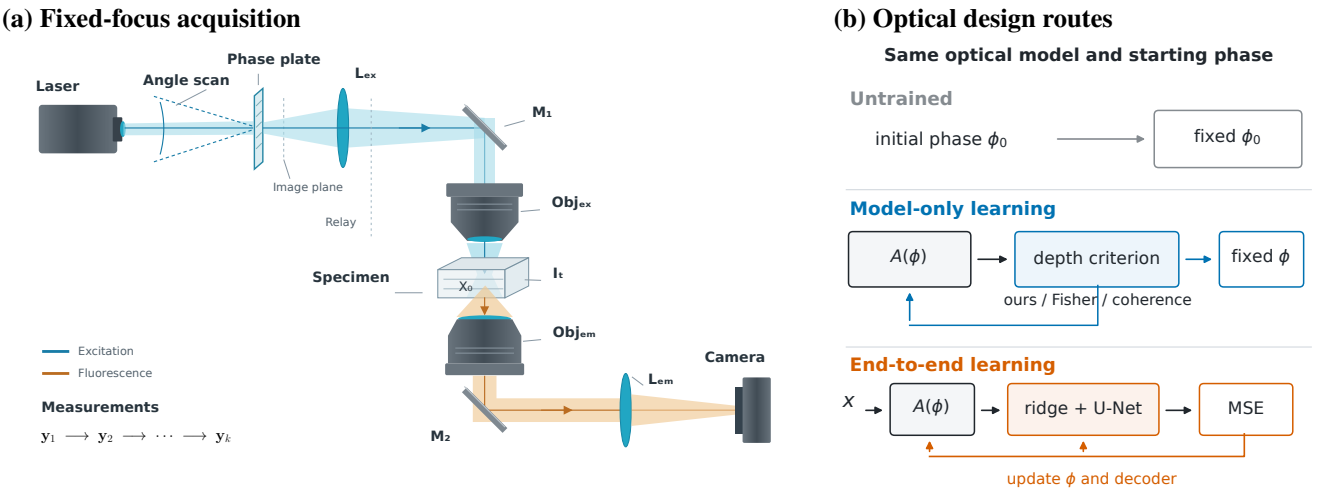


Figure 2. **A shared optical model supports three design routes.** (a) Only the incident angle changes during acquisition; the plate, specimen and detection focus remain stationary. Blue and orange indicate excitation and emission. (b) The untrained phase is retained, optimized using a model-only criterion, or optimized jointly with a decoder on specimen volumes. Each route yields a distinct fixed plate for reconstruction. Shared optics do not imply equal training budgets.

generates an emission source proportional to $X_0 I_t$. Laterally shift-invariant detection sums these contributions incoherently after emission blur and camera-pixel averaging [26]. We compute excitation with a vectorial Richards–Wolf relay to account for high NA [13, 19]. With a piecewise-constant reconstruction basis, the noiseless model is

$$\bar{\mathbf{y}} = \mathsf{A}_\phi \mathbf{x}, \qquad \mathsf{A}_\phi = \mathsf{B}_\phi \mathsf{U}, \tag{1}$$

where B_ϕ acts on a finer optical grid and U replicates each coefficient over its fine-grid cells. This restriction integrates within-voxel optical variation. Measurements are generated from fine-grid specimens; the remaining sub-voxel residual and full image-formation model are specified in the supplement. A column of A_ϕ is the measurement signature of one voxel, stacked across illumination angles and camera pixels.

Decoder-free plate learning. Depths at the same lateral position have overlapping detection responses. We target their excitation-dependent differences by comparing these columns. Let $\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z}$ denote a signature restricted to sampled angles $\mathcal{T}$, centered over its angle–pixel entries and normalized to unit norm. We minimize

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r},\mathcal{T}} \left[\frac{\sum_{z \neq z'} w_{zz'} \left((\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z})^\top \mathbf{a}^{\mathcal{T}}_{\mathbf{r},z'} \right)^2}{\sum_{z \neq z'} w_{zz'}} \right]. \tag{2}$$

Selected reflected-depth pairs receive twice the weight of other pairs, emphasizing similar detection responses; the exact indices are given in the supplement. Squaring penalizes correlation of either sign. Normalization prevents uniform optical-power scaling from directly reducing the loss; excitation dose is matched separately during evaluation. Gradients pass through the excitation, vectorial relay and detector integration, without specimen examples or a reconstruction solver.

Why compare depth signatures? With uniform weights, the unnormalized sum is the off-diagonal energy $\|\mathsf{G} - \mathsf{I}\|_F^2$ of the unit-norm signature Gram matrix. In an idealized single-column Gaussian model, this energy appears in the leading departure of mutual information and linear-MMSE error from orthogonality. This local interpretation motivates the objective; the full volumetric and cross-method gains are evaluated empirically. The derivation and assumptions are in the supplement.

Other model-only criteria. We also optimize a smoothed worst-case coherence objective, $\mathbb{E}[\beta^{-1} \log \sum_{z \neq z'} \exp(\beta |\rho_{zz'}|)]$, with $\rho_{zz'} = \mathbf{a}_z^\top \mathbf{a}_{z'}$, and a local depth Fisher-information surrogate, $-\mathbb{E}_z \log \mathcal{I}_z$, with emitter brightness treated as a nuisance parameter [6, 21, 25]. Both use the same optical parameterization, initial phase and sampled positions/angles as our criterion. Their exact definitions and matched optimization settings are in the supplement.

End-to-end counterpart. Using the same optics and initial phase, we jointly optimize the plate and a decoder on neuronal volumes. A regularized linear solve provides the network input:

$$\begin{aligned} \mathbf{z}_\lambda &= \mathbf{m} + \mathsf{A}_\phi^\top (\mathsf{A}_\phi \mathsf{A}_\phi^\top + \lambda \mathsf{I})^{-1} (\mathbf{y} - \mathsf{A}_\phi \mathbf{m}), \\ \hat{\mathbf{x}} &= \mathcal{N}_\theta(\mathbf{z}_\lambda), \end{aligned} \tag{3}$$

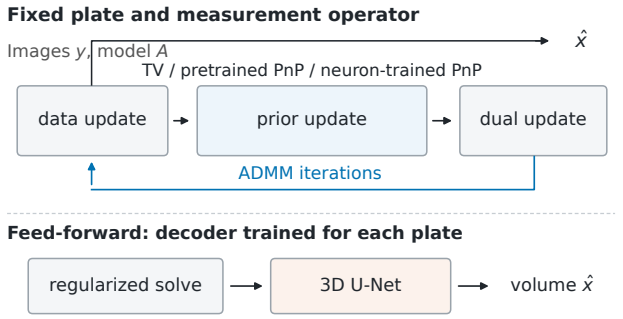


Figure 3. **Reconstruction is evaluated after the plate is fixed.** TV and PnP use ADMM with different prior updates. The feed-forward route uses a regularized solve followed by a 3D U-Net whose weights are trained separately for each plate. Full updates are in the supplement.

where $\mathbf{m}$ is a constant volume at the mean training intensity, $\lambda > 0$ is learned, and $\mathcal{N}_\theta$ is a residual 3D U-Net. Training minimizes reconstruction MSE and updates both ϕ and the decoder parameters through the physical model. The measurement-space system is solved directly. We then fix the resulting phase and evaluate it with all reconstruction methods. Training stages, validation selection and gradient checks are detailed in the supplement.

3.1. Reconstruction with fixed optics

We evaluate four methods with the corresponding operator A_ϕ (Fig. 3). TV combines a quadratic measurement term with isotropic 3D total variation and a $[0, 1]$ box constraint [20]. PnP uses denoisers as implicit priors in ADMM [10, 22]. Following multi-slice fusion [12], pretrained PnP applies a natural-image DnCNN independently to slices along three orientations of one shared volume [28]. Neuron-trained PnP replaces that network with a neuron-trained 2D denoiser and adds a 3D denoising branch.

TV and PnP share a quadratic data term, with different splitting and prior updates. The PnP denoiser weights remain fixed across optical encoders; scalar parameters are selected separately on validation volumes. The fourth method uses Eq. (3) with the phase fixed and a network fine-tuned separately for each plate. Its architecture is shared, but its learned weights are not. Only the end-to-end optical-design route includes a decoder during plate training. Solver updates, constraints and network training are specified in the supplement.

4. Experimental setup

Simulation settings. We simulate linear fluorescence in a homogeneous, non-scattering medium. Excitation and emission use a common wavelength of $532\,\mathrm{nm}$ as a modeling approximation, with numerical apertures of 0.8 and 0.1, respectively. The phase plate has $0.32\,\mu\mathrm{m}$ features and lies $6.6\,\mu\mathrm{m}$ before the intermediate illumination image plane in specimen-equivalent coordinates. Fields use $0.08\,\mu\mathrm{m}$ lateral and $0.32\,\mu\mathrm{m}$ axial sampling, binned to a $0.32\,\mu\mathrm{m}$ analysis grid. Measurements are generated on this grid before averaging the reference into $0.64\,\mu\mathrm{m}$ reconstruction voxels, retaining within-voxel specimen variation.

Volumes and compression. A reconstruction has $16 \times 48 \times 48$ voxels ($N = 36{,}864$), covering $30.72 \times 30.72 \times 10.24\,\mu\mathrm{m}^3$ (lateral $\times$ lateral $\times$ axial). Each camera image contains 12×12 samples. Acquiring $k = 256$, 64 or 32 directions gives measurement compression $C = N/(144k) = 1$, 4 or 8. For $C > 1$, five randomly drawn direction subsets are paired across encoders.

Data and validation. References are public two-photon stacks of mouse pyramidal neurons used in DEEP^2 [23], min–max normalized and resampled to the analysis grid. Six cells from three animals contribute 60 test crops: two per cell selected by reference structure before reconstruction, and eight sampled at random after reconstruction parameters were fixed. Other full-field studies use the 12 prespecified crops unless stated otherwise. Neuronal denoisers and supervised optical/decoder training use 459 crops from a fourth animal. The supervised models use 48 validation crops from a held-out stack of that animal. TV/PnP scalar parameters are selected separately for each encoder and acquisition condition on two vascular crops from another animal. Test volumes are excluded from training, selection and tuning.

Optical design controls. Our criterion, depth Fisher information and smoothed worst-case coherence share the phase parameterization, initialization, sampled positions/angles and 1000-step Adam budget. The end-to-end counterpart uses the same optics and initial phase, with a separate supervised training schedule. Its decoder and those for the untrained and model-only plates are fine-tuned for the same additional number of steps; their earlier training histories differ. These comparisons match the acquisition and physical model, not total training cost. Additional controls include random phase realizations, defocused binary masks, a plate-free scan and an in-focus DMD; full protocols are in the supplement.

Photon budget. All encoders are normalized to the same mean excitation over the specimen and all 256 directions. With k acquired images, camera counts follow $n_i \sim \mathrm{Poisson}(\kappa_0\, 256\bar{y}_i/k)$, with normalized data $y_i = n_i/(\kappa_0\, 256/k)$ for reconstruction. For each specimen, κ_0 is set using the untrained initialization to give 1000 expected photons per camera pixel per image at the mean signal for

$k = 256$, then shared across encoders. This matches deposited dose for the full scan and in expectation over direction subsets; we call it *matched nominal dose*. Detected counts can differ. Fewer images receive longer exposure per image. Apertures, background/read noise and reduced-dose conditions are specified in the supplement.

Metrics and statistics. Our primary metric compares full axial intensity profiles using one-dimensional Wasserstein-1 distance [17]. For each reference-signal column c, normalize its reference and nonnegative reconstructed profiles, with CDFs F_c and $\hat{F}_c$. Then

$$E_{\mathrm{W1}} = \sum_c \pi_c \, \Delta z \sum_{i=0}^{Z-2} |F_c(i) - \hat{F}_c(i)|, \tag{4}$$

where $Z = 16$, $\Delta z = 0.64\,\mu\mathrm{m}$ and π_c is the normalized reference column brightness. Columns must exceed 10 % of the maximum reference brightness. Centroid error uses the same columns and weights. PSNR and foreground SSIM complement these normalized axial metrics with intensity and image-quality assessment. We average draws within each crop and report paired changes in crop means. Confidence intervals resample the six cells with their crops intact (10,000 replicates), accounting for within-cell dependence. Phantom intervals resample phantom cases after averaging draws.

Axial structure tests. Crossing filaments use three lateral geometries, two axial centers and five measurement draws. Success requires matched axial peaks in columns carrying at least half the reference crossing signal, together with correct ordering of the arms. The smallest resolved gap must achieve 80 % success at each center and all larger tested gaps. Separately, measurement discrimination is assessed against merged-depth alternatives preserving the reference projection and column centroids. These prescribed alternatives do not establish uniqueness among all possible volumes. Definitions and selection rules are in the supplement.

5. Results

5.1. Depth discrimination before reconstruction

The untrained defocused plate already creates depth-dependent excitation. Learning makes this code more selective (Fig. 4a): strong responses occupy different axial regions as the incident angle changes. Over the interior field, mean absolute depth-signature correlation falls from 0.845 to 0.480; the end-to-end plate reaches 0.628 (Fig. 4b). These correlations use the full detector-integrated signatures, rather than illumination intensity alone.

We next compare each neuronal reference with a prescribed merged-depth alternative that preserves its projection and column centroids. With global brightness fitted out, our plate increases Poisson measurement divergence by a geometric mean factor of 10.0 relative to the untrained plate (95 % CI: 8.7–11.8), and 2.0 relative to the end-to-end plate (1.9–2.1). Both increases hold for all 59 eligible crops after averaging direction subsets (Fig. 4c). This tests discrimination of specific axial alternatives, independently of a reconstruction prior; it does not establish unique recovery of arbitrary volumes.

5.2. One learned plate, four reconstruction methods

Our plate lowers mean axial-profile W1 by 41 % with TV, 50 % with pretrained PnP, 28 % with neuron-trained PnP and 27 % with the feed-forward decoder, relative to its untrained initialization (Fig. 1b, Tab. 1). Paired cell-bootstrap intervals exclude zero for all four comparisons. W1 improves on all 60 crops under each TV/PnP method. The improvements persist on the prespecified and later randomly selected crops, and a second optical sampling seed gives similar results from the same phase initialization.

Centroid errors also decrease, by 42 %, 54 % and 30 % under TV and the two PnP methods, and PSNR increases by 1.4–1.8 dB across the four methods. Thus, profile recovery improves alongside mean depth placement and image quality. With a non-learned TV prior, our plate also gives lower W1 than the untrained plate with neuron-trained PnP (0.218 versus 0.271 μm).

Figure 5 shows reduced axial smearing along neuronal branches. Pretrained PnP, which has the largest mean gain, is shown; the selected crops have smaller W1 gains than the corresponding test-set median. The supplement includes all-prior views and a weak feature missed by both plates. Additional random-phase realizations place the common initialization within the range of the tested untrained plates. Defocused amplitude masks remain competitive with untrained phase plates in centroid error; their full comparison is reported separately.

5.3. Comparison of model-only objectives

The criteria target different properties: Fisher information measures local depth sensitivity, smoothed worst-case coherence emphasizes the most similar pairs, and our loss penalizes correlations across all depth pairs. All improve upon the untrained plate (Tab. 1). Our criterion gives 12–26 % lower W1 than smoothed worst-case coherence and 10–21 % lower than the depth Fisher-information surrogate across the four reconstruction methods. Paired cell-bootstrap intervals exclude zero in each primary comparison. These results compare the stated implementations under matched sampling and optimization budgets, not the best possible design for each objective. The TV/PnP gains persist with a second optical sampling seed on the same 60 crops; full results are in the supplement. On the 12 prespecified crops, an uncentered squared-correlation variant gives nearly identical

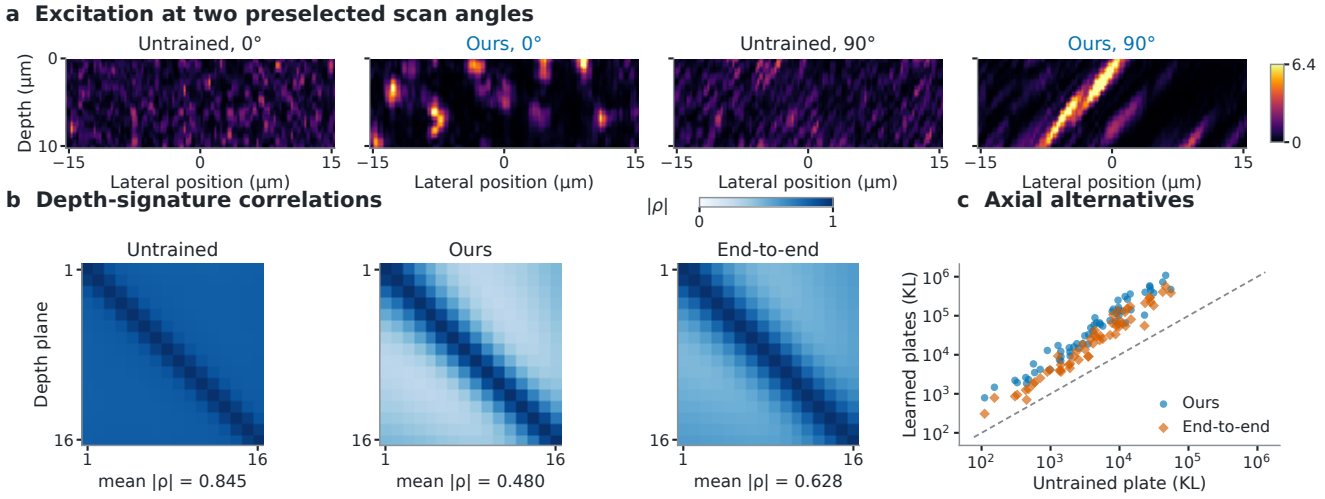


Figure 4. **Optical learning changes the depth code before reconstruction.** (a) Actual excitation sections at two scan angles selected by index, with a shared linear intensity scale relative to the untrained mean. Coordinates span the specimen volume. (b) Absolute correlations between depth-signature pairs, averaged over 44×44 interior positions using the same 32 directions. (c) Poisson measurement divergence from merged-depth alternatives for 59 eligible neuronal crops. Points average the five direction subsets; the diagonal indicates equal divergence.

Reconstruction	Untrained	Depth Fisher	Worst-case coherence	Ours	End-to-end
TV	0.371	0.251	0.264	0.218	0.222
Pretrained PnP	0.579	0.367	0.390	0.290	0.281
Neuron-trained PnP	0.271	0.218	0.223	0.195	0.189
Feed-forward	0.366	0.314	0.314	0.265	0.234

Table 1. **Optical design evaluated across four reconstruction methods.** Mean axial-profile W1 (μm) on simulated measurements at $C = 8$ and matched nominal dose: 60 crops from six cells and three animals, averaging five direction subsets per crop. Model-only criteria use sampling seed 0 and matched optimization settings. The feed-forward network is fine-tuned separately for each plate. The untrained plate is the common phase initialization. Paired intervals and intensity metrics are in the supplement.

recovery under neuron-trained PnP (0.207 versus 0.206 μm), so centering is not an established source of the advantage.

5.4. Comparison with joint optical–decoder training

The end-to-end plate improves substantially over the untrained plate and also transfers to reconstruction methods outside its training loop (Tab. 1). Relative to ours, its mean W1 is 1.9 % higher under TV, 3.0 % lower under pretrained PnP and 3.2 % lower under neuron-trained PnP. The paired intervals exclude zero in these directions; these small differences should not be read as established equivalence.

When each plate uses its separately fine-tuned feed-forward decoder, the end-to-end system has 12 % lower W1 (0.234 versus 0.265 μm) and 0.48 dB higher PSNR than ours. This comparison includes different decoder training histories; the TV/PnP comparisons hold the priors fixed. Model-only optical learning attains close axial recovery under the independent TV/PnP methods without requiring specimen examples or a decoder during optical training. The comparison concerns this architecture and training protocol, rather than all end-to-end designs.

5.5. Recovery of overlapping structures

In the controlled crossing task, the smallest gap meeting the resolution-and-ordering criterion falls from 3.84 to 2.56 μm with our plate under TV and pretrained PnP (Fig. 6). It remains 3.84 μm under neuron-trained PnP despite improved neuronal W1. The zero-gap control shows no false splitting. At full dose, crossing-success differences between the two learned plates are not established under either evaluated method.

Reduced dose exposes a difference in this task. At one tenth of the nominal dose, our plate has 17 percentage points higher success under TV (95 % CI: 6–30) and 15 points higher under pretrained PnP (2–30), pooling the 18 phantom cases with gaps of 1.28–2.56 μm. At the same dose on neuronal crops, W1 differences between the two learned

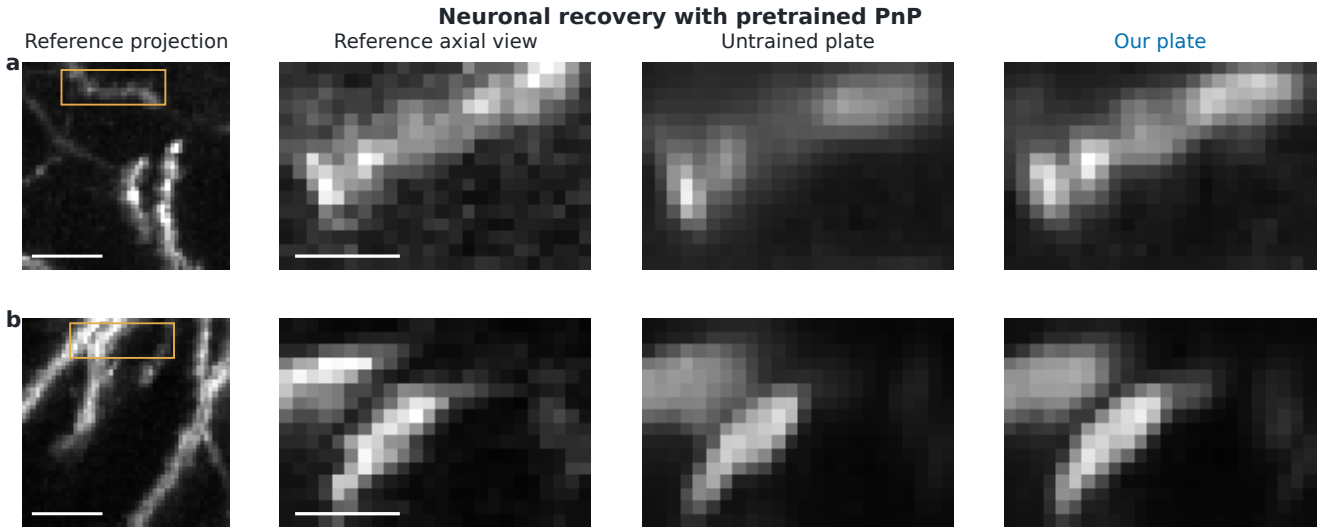


Figure 5. **Less axial smearing in neuronal reconstructions.** Two reference-selected crops from different animals, reconstructed with pretrained PnP at $C = 8$. Left: reference projections with the displayed axial bands outlined. The remaining columns show reference, untrained-plate and learned-plate band views at true aspect ratio, with depth downward. Each row uses one linear intensity scale fixed from its reference. Crop and band selection do not use reconstruction gains. Bars: $10\,\mu$m in projections, $5\,\mu$m in axial views. All-prior views and additional depth slabs are in the supplement.

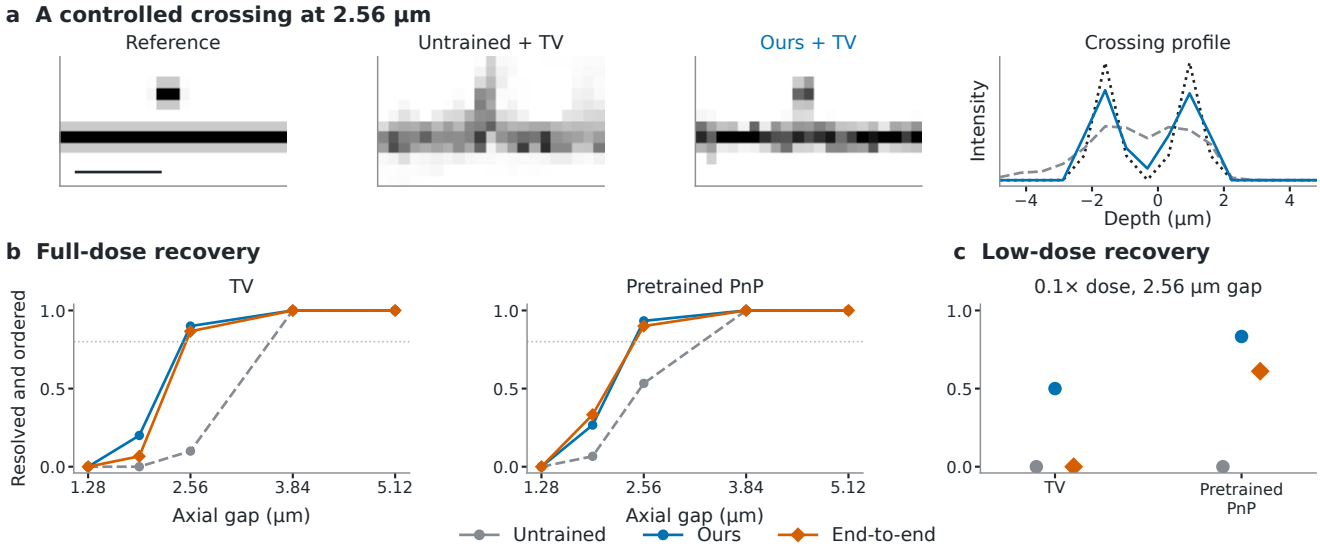


Figure 6. **Controlled structure recovery depends on the code, prior and dose.** (a) Matched sections and profiles through one crossing at our plate's TV threshold gap; the first fixed geometry, center and draw are shown, with a shared intensity scale. Bar: $5\,\mu$m. (b) Full-dose success rates pool the two axial centers; the 80 % criterion is applied to each center separately. (c) At one tenth of the dose, our plate succeeds more often at the displayed gap. Full gap curves and paired phantom-bootstrap intervals are in the supplement.

plates are not established. Larger-gap three-filament tests and weak-branch controls are in the supplement.

5.6. Acquisition budgets and calibration

With 32 images, our plate gives lower W1 than the untrained plate with 256 images under TV (0.196 versus $0.249\,\mu$m) and pretrained PnP (0.277 versus $0.413\,\mu$m). Neuron-trained PnP gives close means (0.206 versus $0.203\,\mu$m). These comparisons use the same nominal dose, with longer exposure per image for the shorter scan. At 0.3 times dose, our plate reduces W1 by 22–39 % relative to the untrained plate at full dose. Gains also persist with background and read noise; full curves are in the supplement.

Calibration is important for preserving these gains. Known plate-distance offsets of $\pm 0.5\,\mu\mathrm{m}$ retain 27–28 % W1 reductions under neuron-trained PnP after updating the operator and revalidating scalar parameters, without retraining the plate. Unmodeled distance or lateral offsets can reverse the advantage: a one-feature lateral shift makes our plate 11 % worse than the untrained control (95 % CI: 2–23). Learned encoding therefore requires registration of the realized optics.

6. Discussion and limitations

Optical design before decoder choice. Learning from physical depth signatures produces an encoder useful under four reconstruction configurations. The tested joint design also transfers, with slightly lower W1 under both PnP methods and a larger advantage with its own decoder. Model-only design provides substantial axial-recovery gains without requiring specimen examples or a decoder during optical training, while joint design can exploit supervised feedback. The comparison makes the training requirements and resulting recovery of both approaches explicit.

Measurement discrimination and structure. Our plate has lower depth correlations and greater separation from prescribed merged-depth alternatives than the end-to-end plate, yet these diagnostics do not yield a universal ranking of neuronal reconstruction error. They target particular ambiguities. Controlled crossings provide a complementary consequence: learning reduces the qualifying gap under two priors, and model-only design improves low-dose crossing success over the tested joint design. These tests concern axial profiles and controlled separation, not general neuronal connectivity.

Physical implementation. All results are numerical, using linear fluorescence in a homogeneous medium with idealized optics and a common excitation/emission wavelength. Fine-grid measurement generation retains specimen approximation error but shares optical physics with reconstruction. A measured system must characterize its realized excitation and detection responses. The alignment study supports updating the operator for known perturbations, while unmodeled registration can erase the gain.

Scope of the comparison. The test set contains six cells from three animals, with dependent crops and approximate cell-bootstrap intervals. Two optical sampling seeds share one phase initialization. We test specific coherence and Fisher surrogates and one jointly trained architecture, with different overall training histories and an MSE-trained supervised decoder. The results establish the reported comparisons, rather than an optimum over all objectives, initializations or reconstruction methods.

7. Conclusion

We learn depth-encoding illumination from the optical model alone, without specimen examples or a reconstruction decoder. In fixed-focus fluorescence simulations, the same phase plate improves axial-profile recovery across four reconstruction methods and gives competitive results against a jointly trained plate. Joint training retains an advantage with its own decoder, while the model-only criterion improves the tested low-dose crossing task. These results show how targeting physical depth responses can improve axial recovery while leaving reconstruction choice outside optical training.

LLM Usage Statement

OpenAI ChatGPT/Codex and Anthropic Claude assisted with experimental planning and code, manuscript and supplement organization, drafting and editing, literature checks, numerical audits of simulation outputs, and figure/table preparation. The authors retain responsibility for the methods, results, citations and final manuscript.

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